Nimeth Nimdinu

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Moratuwa, Sri Lanka



PROFESSIONAL SUMMARY

Results-driven final-year IT undergraduate at the University of Moratuwa, specializing in Artificial Intelligence and Machine Learning, with hands-on industry experience in developing AI-powered applications and intelligent systems. Skilled in machine learning, deep learning, data-driven problem solving, and scalable software development.

CORE SKILLS

Programming & ML Frameworks: Python, PyTorch, TensorFlow, Scikit-learn, Transformers, LangChain, LangGraph
Generative AI & LLMs: LLM Fine-tuning (LoRA, PEFT), RAG, Graph RAG, Multi-Agent Systems (A2A), MCP Server
Cloud DevOps: AWS (Serverless, S3), Azure, Azure DevOps, CI/CD, Docker, FastAPI, Model Deployment
Vector DBs & Data Storage: Pinecone, Neo4j (Knowledge Graphs), Redis, MongoDB, PostgreSQL, MSSQL
Software System Design: REST APIs, Microservices, Authentication, System Design, SDK Integration
Soft Skills: Problem Solving, Critical Thinking, Client Engagement, Teamwork, Agile Practices, Leadership

EXPERIENCE

Thakshana Technologies Sep 2026 – Present
AI Engineer Colombo, Sri Lanka

- Developing an AI-powered hardware coding assistant to automate the Vivado/VHDL implementation lifecycle, including RTL generation, simulation, synthesis, and verification
- Building a LangGraph-based multi-agent workflow with Ollama for automated design analysis, testing, failure diagnosis, and iterative code repair
- Tech Stack: Python, LangChain, LangGraph, LangSmith, MCP, MLFlow, Fast API, Ollama, Matlab, Streamlit

NovaCodex Oct 2025 – Sep 2026
Associate Software Engineer Moratuwa, Sri Lanka

- Conducting root cause analysis on ticketing system issues to identify recurring bugs, workflow bottlenecks, failures
- Building payment processing platform for event ticketing and product selling with multi-gateway orchestration
- Tech Stack: PyTorch, GNN, TGNN, Fast API, Node.js, Next.js, AWS (Serverless, S3), MongoDB

Creative Software Mar 2025 – Oct 2025
Intern Software Engineer (R&D) Colombo, Sri Lanka

- Added GenBI using RAG to enable natural language queries and LLM-powered automated insights for DORA metrics analytical system for engineering performance, allowing teams to ask metrics questions conversationally
- Built conversational GenUI interactive chat system generating dynamic HTML components from natural language
- Explored LLM-based automation, agent architectures, and AI-driven workflows for research initiatives
- Tech Stack: Python, Langchain, LangGraph, Sementic kernel, PyQt6, n8n, Azure Devops CI/CD , ASP.NET Core

EDUCATION

University of Moratuwa Jun 2022 – Jun 2026
BSc (Hons) in Information Technology Moratuwa, Sri Lanka

- CGPA: 3.78/4.0

Sivali Central College Mar 2018 – Oct 2020
Physical Science Stream - 3A's(Combined maths, Physics, ICT) Ratnapura, Sri Lanka

PROJECTS

Tea Farming Support Assistant | *AI and Fullstack developer* May 2024 – Jul 2024

- Developed a modular agent architecture with a central FastAPI service that classifies user queries using an LLM and routes them to specialized services.
- Disease Service using retrieval-augmented generation (RAG) with Pinecone vector database and web search
- Fertilize Service providing personalized recommendations using weather APIs, tea land data, historical records
- Designed a Help Service for company policy and FAQ retrieval leveraging Pinecone vector database.
- Integrated LangChain and LangGraph for agent orchestration and developed the frontend using Next.js.
- Tech Stack:Python, LangChain, LangGraph, MLFlow, Pinecone, Scikit-learn, Pandas, FastAPI, Next.js, DuckDuckGo

Multi-Agent A2A System | *AI Developer (Ongoing)* July 2026 – Present

- Designed and implemented an autonomous Agent-to-Agent (A2A) communication system enabling multiple AI agents to collaborate and exchange contextual information dynamically.
- Developed modular Host and Remote agent architectures supporting intelligent task delegation, workflow orchestration, and feedback-driven coordination.
- Implemented multi-agent communication pipelines and reasoning workflows using CrewAI, LangGraph, and Google ADK frameworks.
- Built proof-of-concept demonstrations showcasing cross-agent collaboration, contextual decision-making, and autonomous task execution.
- Tech Stack: Python, CrewAI, LangGraph, Google ADK, FastAPI, LangChain

Domain-Adapted Farming LLM | *ML Developer* Jun 2025 – Sep 2025

- Developed a domain-adapted GPT-2 based language model for Sri Lankan tea cultivation using a two-stage fine-tuning pipeline with PEFT techniques.
- Performed domain adaptation on agricultural literature using LoRA (Low-Rank Adaptation) to improve tea farming knowledge representation.
- Implemented supervised fine-tuning (SFT) on question-answer datasets to generate context-aware and task-specific agricultural responses.
- Optimized model training and deployment using 4-bit quantization, adapter merging, and progressive fine-tuning for improved memory efficiency and inference performance.
- Tech Stack: Python, PyTorch, Transformers, Hugging Face, LoRA, Pandas, Ollama

Developer productivity analytical system | *Software Engineer (Intern)* Mar 2025 – Sep 2025

- Implemented developer productivity analytics using DORA and SPACE metrics to track team performance.
- Integrated secure data collection and multi-platform connectivity with GitHub and Azure DevOps APIs using Python.
- Implemented CI/CD pipelines with Azure DevOps and deployments on on-premise Apache Tomcat environments.
- Tech Stack: Python, .Net core, React, TypeScript, Entity Core, MSSQL, Azure, PyQt6, Gradio

Ontology-guided GNN for Semantic Candidate Matching | *Lead Researcher* Mar 2026 – Jul 2026

- Designed a software industry ontology knowledge graph schema to model fine-grained skills and semantic relations.
- Built a GNN over the ontology-guided KG to learn Job and Candidate embeddings and compute embedding similarity.
- Combined ontology-based scoring and GNN similarity to produce matched candidates
- Tech Stack: Python, Neo4j/RDF,PyTorch, PyTorch-Geometric, spaCy, pandas, scikit-learn

ACHIEVEMENTS

HackX - University of Kelaniya Finalist (Top 10)

SLIOT - University of Moratuwa Finalist (Top 10)

CERTIFICATES

Azure Fundamentals (AZ - 900)

AI Agents Using RAG

Gen AI with Transformers

REFERENCES

Laisth Chandrasekara Software Engineering Manager, Creative Software

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